

IITG.AI · SUMMER 2026

ARXIV:2602.11708

Intraday Volatility Forecasting & Adaptive Trend-Following

Jump-BiLSTM + Attention

Best Model

BTC/USDT · 5-min bars · Jan 2023 – Jun 2026 · PyTorch on Kaggle T4

0.4319

QLIKE Score

QLIKE gain +56.7% vs HAR-RV 0.9900

1.838

Sharpe Ratio

Sharpe ratio

+33.24%

Net Return

Return +33.24% vs B&H -35.79%

Strategy: AdaptiveTrend

Project Overview

End-to-end pipeline: raw market data → deep learning forecasting → systematic trading strategy

1

O1 • Data Collection

BTC/USDT + ETH/USDT
5-min OHLCV bars
Jan 2023 – Jun 2026
368,333 bars · KuCoin

2

O2 • Forecasting

3 DL models vs HAR-RV baseline
QUKE loss function
21-feature input set

3

O3 • Strategy Development

3 approaches tested
2 failed (documented)
1 succeeded
Root causes explained

4

O4 • Results

AdaptiveTrend k=4.0
Test: Jan–Jun 2026
Sharpe 1.838
Zero leakage

🔒 **Train:** Jan 2023–Jun 2025 | **Val:** Jul–Dec 2025 | **Test:** Jan–Jun 2026 (run once — strictly no leakage)

Phase 1 • Feature Engineering

21 strictly causal inputs · (batch, 78, 21) tensor · perturbation-verified zero leakage

Corsi (2009) • BN&S 2004 • BNK&S 2010

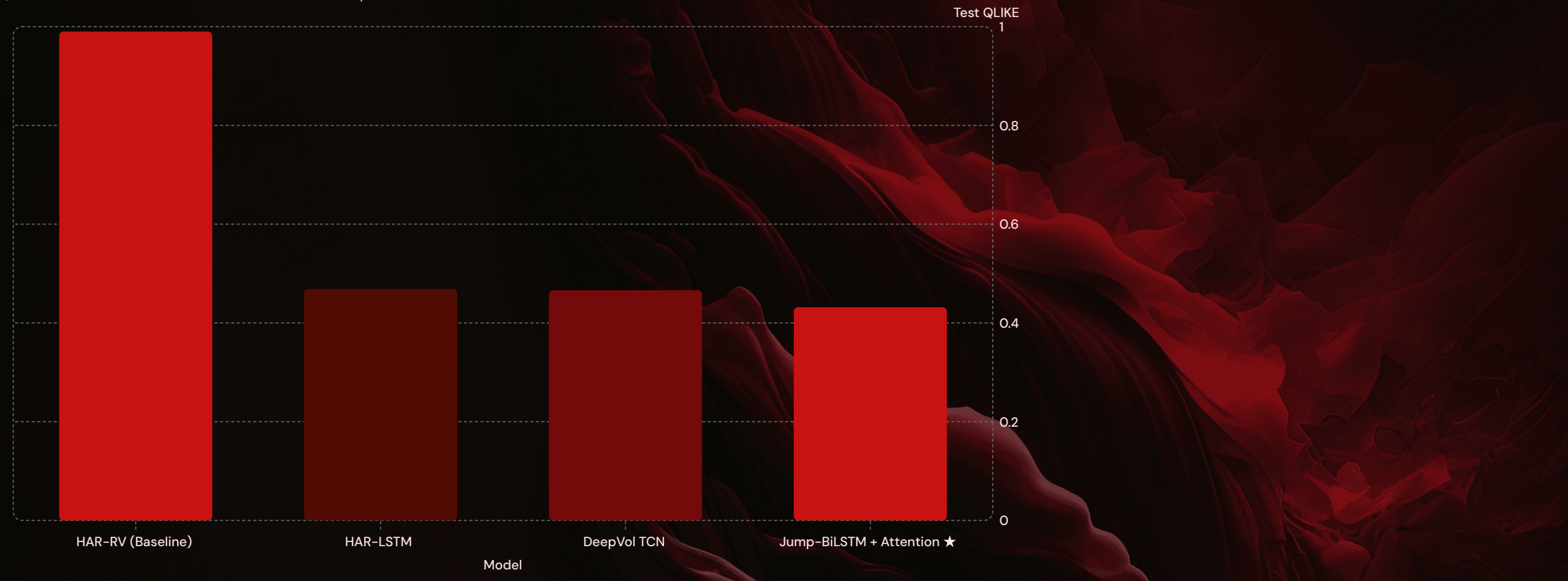
- **RV_1bar** — instantaneous (5-min)
- **RV_continuous** — smooth diffusion part
- **RV_1hr** — 1-hour rolling average (12 bars)
- **RV_pos_jump** — upward spike contribution
- **RV_1day** — daily average (288 bars)
- **RV_neg_jump** — crash contribution
- Captures multi-scale vol memory
- Reconstruction error: 1.08×10^{-19}

Standard • Zhang et al. JFEC 2024

- RSI(14), EMA_10, EMA_50, ATR(14)
- ETH_RV_1bar / 1hr / 1day
- GK_vol — Garman-Klass OHLC estimator
- ETH_log_return — momentum
- BB_width, VWAP_deviation
- DVOL — Deribit BTC implied vol (VIX equivalent)
- sin_time + cos_time (intraday seasonality)
- Cross-asset commonality signal

Phase 1 • Three Models vs HAR-RV

QLIKE loss — lower is better. All three DL models beat the linear baseline by 52–57%.



BASELINE • HAR-RV Corsi (2009) Linear regression on 3 HAR lags. No learning. Our floor to beat. TEST QLIKE: 0.9900	MODEL 1 • HAR-LSTM Zhang et al. 2024 2-layer LSTM · hidden=64 · Dropout 0.2 · Final hidden state · Adam lr=1e-3 · patience=5 TEST QLIKE: 0.4689 (+52.6%)
MODEL 2 • DeepVol TCN Moreno-Pino & Zohren 2024 7-layer dilated TCN · Dilation 2 ¹ · receptive field 254 · Residual blocks · CosineAnnealingLR TEST QLIKE: 0.4661 (+53.1%)	★ BEST • Jump-BiLSTM + Attention BiLSTM · self-attention over 78 bars · Jump features explicit · AdamW · Grad clip 0.35 · patience=10 TEST QLIKE: 0.4319 (+56.7%)

❏ All architectures converge to QLIKE 0.43–0.47 → the feature set is the information ceiling, not model capacity

Jump Decomposition – The Key Feature Innovation

The Problem

HAR–RV squares all returns equally. A \$500 smooth drift and a \$500 flash crash both contribute identically to RV. But their impact on future volatility is totally different.

The Mathematics

Bipower Variation – isolates smooth component:

$$BV = \frac{\pi}{2} \cdot \frac{M}{M-1} \cdot \sum |r_i| \cdot |r_{i+1}|$$

Adjacent returns product – jumps cancel out

Jump variance – positive excess over BV:

$$J = \max\{0, RV^2 - BV\}$$

Large $J \rightarrow$ a jump happened this window

Semivariance – split by direction:

$$RS^+ = \sum r^2 \cdot 1[r > 0] \quad RS^- = \sum r^2 \cdot 1[r < 0]$$

Negative jumps hit harder – Zhang et al. 2024

Reconstruction verified:

$$\|\text{continuous}^2 + \text{pos}^2 + \text{neg}^2 - RV^2\| = 10^{-19}$$

Perfect decomposition – floating point only

The Solution – Three Components

RV_continuous

Smooth diffusion. Price drifted normally. Moderate, predictable future volatility.

RV_pos_jump ↑

Upward spike. Liquidation cascade up. Mild persistence in future vol.

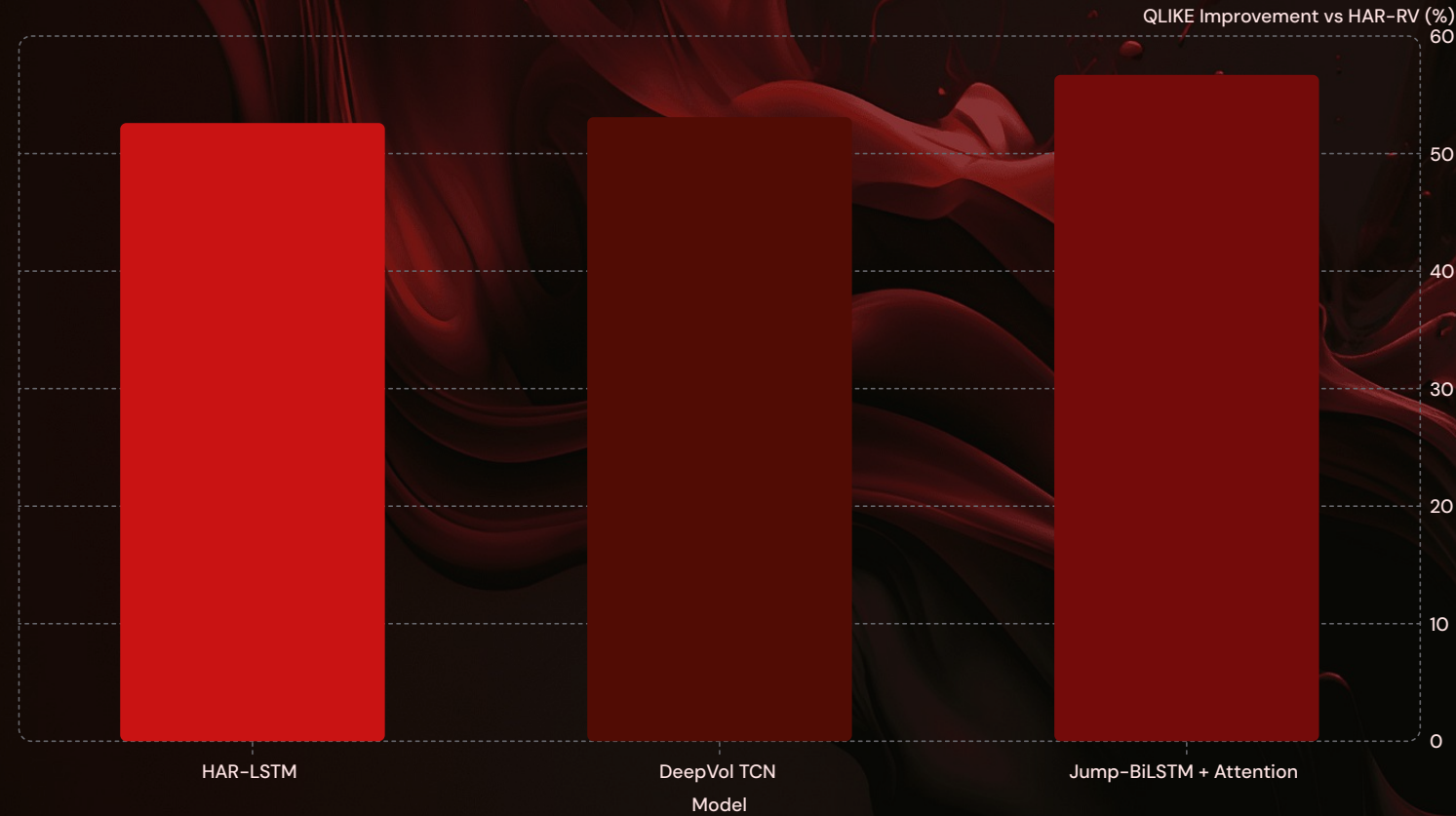
RV_neg_jump ↓

Downward crash. Flash dump / panic. HIGH and persistent future volatility.

Phase 1 • Results

+56.7%

HAR-RV 0.9900 → Jump-BiLSTM 0.4319
The gap between the 3 DL models (0.43–0.47) is smaller than the gap to baseline (0.99) — feature set is the ceiling.
QLIKE improvement over HAR-RV baseline
HAR-LSTM 0.4689 · TCN 0.4661 · BiLSTM 0.4319



Calibrated Prediction Bands

89.9% coverage at k=2.15
Target: 90%
BiLSTM predicted vol is a well-calibrated uncertainty estimate, not just a point forecast. Validated on the held-out test set.

News Sentiment Ablation

< 0.5% QLIKE change
NULL FINDING
648 FinBERT-scored headlines. 99.7% of 5-min bars have zero news. Intraday BTC vol is microstructure, not macro sentiment.

Architecture Convergence

0.43–0.47
Information ceiling
Marginal gains from architecture beyond BiLSTM are negligible. Richer features would improve all models equally.

Phase 2 • Two Approaches Failed First

Documenting failures is part of good research — both teach something fundamental about crypto vs equities.

❑ FAILED

VWAP Mean Reversion

Claimed: Sharpe 3.02 (1H bars, in-sample)

Reality

0/6 months positive in walk-forward

Mean Sharpe across all test months: -35.5

Root Cause

Val period (Jul–Dec 2025) was RANGING. Test period (Jan–Jun 2026) was TRENDING. Mean reversion + trending market = buying every dip that keeps going down.

Lesson

In-sample Sharpe 3.02 was pure regime luck, not real signal. Test set confirms.

❑ FAILED

Moreira-Muir Vol-Scaling

Claimed: Proven strategy in equities (JF 2017)

Reality

Return -33.20% vs B&H -33.12%

Underperformed even buy-and-hold

Root Cause

56 param combos tested: ALL failed OOS. Equity assumption: high vol → bad returns. Crypto reality: high vol = bull runs. BTC's biggest rallies ARE its most volatile. M&M reduces exposure on the BEST days. Structurally incompatible — not a param issue.

Lesson

The vol-return relationship is INVERTED in crypto. Cannot be parameterised away.

AdaptiveTrend • How It Works

arXiv:2602.11708 (Duc Bui, Feb 2026) · adapted for single-asset BTC with BiLSTM-driven trailing stops



Entry – EMA(20) Crossover

Every 6H bar: is BTC above or below its 5-day EMA?

Above EMA → **LONG BTC**

Below EMA → **SHORT BTC**

Always in the market. Never flat.

One Parameter

k = 4.0 — ATR multiplier

Selected on VAL

Locked before TEST

Never revised

Stop – BiLSTM Adaptive Width

$$\text{stop_dist} = k \times \text{ATR}(14) \times \text{vol_factor}$$

$$\text{vol_factor} = \frac{\text{pred_vol}}{\text{median_train_pred_vol}}$$

High pred_vol → wider stop → ride volatile rallies

Low pred_vol → tighter stop → lock gains faster

Exit & Flip

Stop ratchets only in profit direction (never widens)

Stop hit → exit, immediately enter opposite direction

k = 4.0 selected on VAL · locked · never changed

6H bars = **72 × 5-min**

EMA period: **20 bars = 5d**

ATR period: **14 bars = 3.5d**

vol_factor clip: **[0.5, 3.0]**

Fee: **4bps taker**

1 free parameter. Clean. Zero leakage.

Phase 2 • Test Results – Jan-Jun 2026

BTC fell \$109k → \$73k during test period. k=4.0 locked from val. Run exactly once.

1.838

Sharpe Ratio
B&H: -1.935

-20.68%

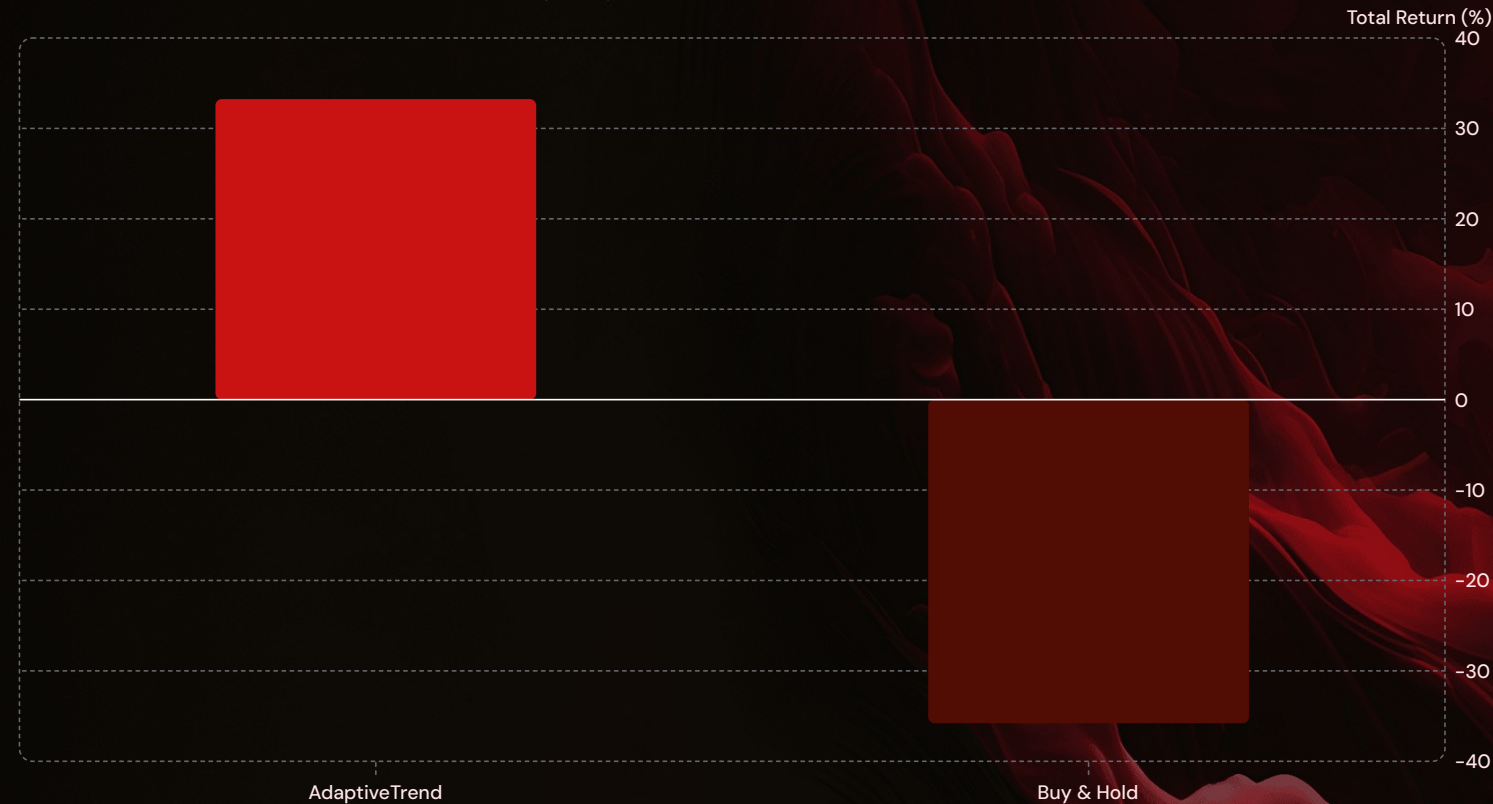
Max Drawdown
B&H: -42.21%

+33.24%

Total Return
B&H: -35.79%

+\$3,324

Net PnL (\$10k)
B&H: -\$3,579



Metric	AdaptiveTrend	Buy & Hold
Time in market	100%	100%
Trades / Avg hold	16 / ~11 days	-
Win rate	50%	-
Total fees paid	~\$50	-
Sharpe improvement vs B&H	+3.773	-

Monthly Breakdown



Honest Assessment · What We Can and Cannot Claim

Bootstrap Confidence Interval (n=1,000)

Sharpe 1.838

✓ CAN CLAIM

- Economically significant (+69pp return differential)
- Sharpe 3.77 points above buy-and-hold
- April loss mechanistically explained (whipsaw)

❌ CANNOT CLAIM

- Statistical significance at 95% confidence
- 95% CI: [-1.109, +4.639]
- $P(\text{Sharpe} > 0) = 91.1\%$
- Sharpe will be positive in next period
- Strategy works across all market regimes

❏ *CI includes zero · 16 trades · 6 months · wide interval is expected and honest*

Key Limitations

1

Sample Size

16 trades, 6 months. Walk-forward over 24–36 months needed for statistical proof.

2

Single Asset

Paper used 150+ coins. Diversification reduces DD to -12.7% vs our -20.68%.

3

No Slippage/Funding

Short funding rates on perps not modelled (0.1–0.3%/day in bull markets).

4

BiLSTM Contribution Modest

+0.019 Sharpe. Trend signal drives returns; BiLSTM improves stop calibration.

Conclusions

1. BiLSTM beats HAR-RV by 56.7% QLIKE

All DL models improve 52–57%. Feature set is the binding constraint.

Attention + jump decomposition = the extra gain.

2. News sentiment: clean null finding

<0.5% QLIKE change across all models. 99.7% of 5-min bars have no news.

Intraday vol = microstructure, not macro.

3. Equity strategies fail structurally on crypto

VWAP MR: 0/6 months OOS. Moreira-Muir: vol-return relationship is INVERTED in crypto. Fundamental incompatibility.

4. AdaptiveTrend: Sharpe 1.838 vs B&H -1.935

+33.24% vs -35.79%. Max DD -20.68% vs -42.21%. 1 parameter, zero leakage. Economically significant.

In trending crypto markets – follow momentum with volatility-adaptive exits. Don't fight the trend. Don't scale by volatility.